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Class: CS 478
Section: 001

What did you build?

We built an LLM-powered agent that enhances Google Scholar by adding a new citation converter feature. This feature allows users to convert academic citations from one format to another. For example, from MLA to APA. It is implemented as a plugin for Gentopia and can be integrated seamlessly with other tools to perform citation formatting tasks. Users need to input the citation they want to convert and specify the desired citation format, and the agent transforms it accordingly. The core functionality is powered by OpenAI's GPT models, which interpret user inputs and provide correctly formatted citation outputs.

Why is it important/useful/interesting?

The citation converter is important because it simplifies the time-consuming task of formatting citations, which is critical for students and researchers. Manually formatting citations is time-consuming and could also lead to errors. As students, there were many instances in the past where we lost crucial points in assignments due to incorrect citations. Also, many existing tools are either limited in scope or not integrated into widely used platforms like Google Scholar. By using an LLM agent, the feature not only confirms accuracy but also provides flexibility, making citation management faster and less stressful. This enhancement can greatly improve productivity and streamline the research process without worrying about dedicating time for citation conversions. Integrating this into Google Scholar makes it much easier for users to convert their citation in Google Scholar itself, instead of relying on third party sources where the results can vary.

Describe your work

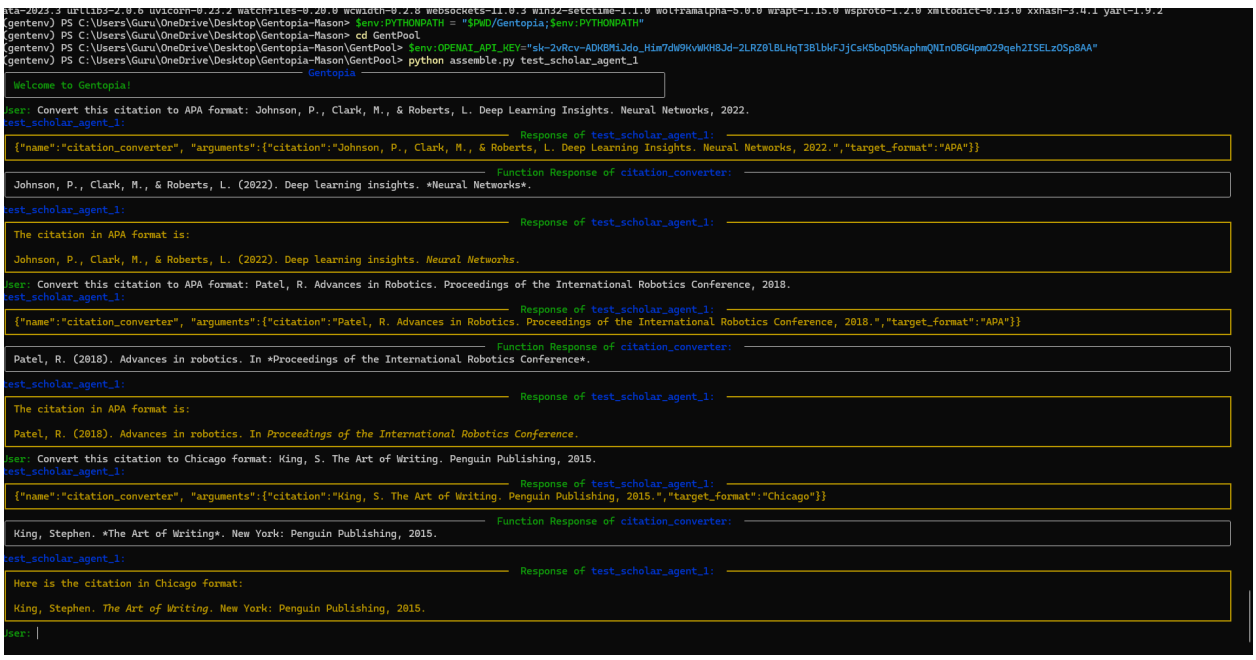
The Citation Converter is integrated into Google Scholar AI as a feature. Users provide the original citation to the chatbot in any legal recognized format along with their desired target format. The LLM processes the input and recognizes the structure of the provided citation. It then converts into the user's desired format and follows the rules the desired format requires. The processing also shows whether the agent takes in the right information under the right criteria. It handles incorrect or incomplete citations by prompting users for clarification.
Once it is processed, the formatted citation is displayed to the user in the specified format including with the correct font.

The agent is built within Gentopia Mason, which provides a modular structure for tool development and integration. The CitationConverter plugin is a custom tool implemented as a subclass of BaseTool. It uses the OpenAIGPTClient to interact with OpenAI's API for processing citation conversion prompts. The CitationConverter defines an args_schema that validates user

inputs which is the citation text and target format. The agent is part of a multi-tool agent in Gentopia and interacts with other plugins when needed. We came up with several prompts to instruct the agent on how to handle different citation formats. This way we trained the agent with multiple citation examples along with a few edge cases.

Example Use Case:
Input:
"Convert this citation from APA to MLA: *Smith, J. (2020). The role of AI in modern education. Journal of AI Research, 15(2), 45-60.*"
Output:
"*Smith, John. “The Role of AI in Modern Education.” Journal of AI Research, vol. 15, no. 2, 2020, pp. 45–60.*"

Figure:



Discussion.

As we worked on this project, we learned a lot of new things along with facing a few challenges. A good thing while working on this project was, the agent showed consistent accuracy in recognizing citation components such as the author, title, etc and applied the correct rules while converting the citation to the user desired citation. The major challenges we faced while working on this project was handling edge cases. These cases included missing citation components or trying to convert it to non-standard citations. Converting non-standard citations would require additional prompt refinement. Initially we also had an issue with getting the wrong citation format which is opposite to what the user desired for. After taking a close look into our

code, we figured out the issue that caused the error. A surprising aspect of this agent was adding redundant information to the citation which over complicated a simple citation. The model GPT-4o-mini was truly helpful in bringing this feature to life. It didn't cause any major disturbances when our code was working perfectly. When our code had issues, the GPT model did struggle with providing the desired outputs. If we were given more time, we would be able to test this agent with more edge cases and come up with solutions when few citation components are missing. We would also consider other non-standard citations and try to include them in this feature which would allow us to cater towards a broader audience. Another idea we had was to gather all the citations and their conversions the user wanted and export them as a word file or a PDF to their local files, which would make it easier for them to access.

Windows 64bit Proofs:

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Gentopia
Welcome to Gentopia!

User: Convert this citation to APA format: Johnson, P., Clark, M., & Roberts, L. Deep Learning Insights. Neural Networks, 2022.
test_scholar_agent_1:
Response of test_scholar_agent_1:
{"name":"citation_converter", "arguments":{"citation":"Johnson, P., Clark, M., & Roberts, L. Deep Learning Insights. Neural Networks, 2022.","target_format":"APA"}}

Function Response of citation_converter:
Johnson, P., Clark, M., & Roberts, L. (2022). Deep learning insights. *Neural Networks*.

test_scholar_agent_1:
Response of test_scholar_agent_1:
The citation in APA format is:

Johnson, P., Clark, M., & Roberts, L. (2022). Deep learning insights. Neural Networks.

User: Convert this citation to APA format: Patel, R. Advances in Robotics. Proceedings of the International Robotics Conference, 2018.
test_scholar_agent_1:
Response of test_scholar_agent_1:
{"name":"citation_converter", "arguments":{"citation":"Patel, R. Advances in Robotics. Proceedings of the International Robotics Conference, 2018.","target_format":"APA"}}

Function Response of citation_converter:
Patel, R. (2018). Advances in robotics. In *Proceedings of the International Robotics Conference*.

test_scholar_agent_1:
Response of test_scholar_agent_1:
The citation in APA format is:

Patel, R. (2018). Advances in robotics. In Proceedings of the International Robotics Conference.

User: Convert this citation to Chicago format: King, S. The Art of Writing. Penguin Publishing, 2015.
test_scholar_agent_1:
Response of test_scholar_agent_1:
{"name":"citation_converter", "arguments":{"citation":"King, S. The Art of Writing. Penguin Publishing, 2015.","target_format":"Chicago"}}

Function Response of citation_converter:
King, Stephen. *The Art of Writing*. New York: Penguin Publishing, 2015.

test_scholar_agent_1:
Response of test_scholar_agent_1:
Here is the citation in Chicago format:

King, Stephen. The Art of Writing. New York: Penguin Publishing, 2015.

User:
Bye!
(gentenv) PS C:\Users\Guru\OneDrive\Desktop\Gentopia-Mason\GentPool> [System.Environment]::Is64BitOperatingSystem
True
(gentenv) PS C:\Users\Guru\OneDrive\Desktop\Gentopia-Mason\GentPool>
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      bursts, leaving plenty of time for exploration.
    • Activities: Visit rice terraces, enjoy spa treatments, and explore cultural temples.
  2 Costa Rica
    • Why visit: The rainy season in Costa Rica is ideal for experiencing vibrant wildlife and
      lush rainforests. The Pacific coast is particularly enjoyable during this time.
    • Activities: Go hiking in national parks, visit waterfalls, and enjoy hot springs.
  3 New Orleans, USA
    • Why visit: New Orleans has a unique charm, and while it does have rain, the city's vibrant
      culture, music, and cuisine can be enjoyed regardless of the weather.
    • Activities: Explore museums, enjoy jazz music, and indulge in local food.
  4 Vietnam (Hanoi & Halong Bay)
    • Why visit: The rainy season can be a great time to visit northern Vietnam, where the rain
      enhances the scenery. The temperature is also cooler.
    • Activities: Cruise in Halong Bay, explore the Old Quarter in Hanoi.
  5 Amazon Rainforest (Brazil)
    • Why visit: Visiting during the rainy season means the rainforest is at its most vibrant.
      Wildlife is abundant, and the river systems are full.
    • Activities: Guided jungle tours, boat rides, and wildlife spotting.

                                  Final Answer

 Here are some recommended travel destinations during the rainy season:

  1 Bali, Indonesia
    • Tropical beauty, cultural experiences, and fewer crowds.
  2 Costa Rica
    • Lush rainforests, diverse wildlife, and adventure activities.
  3 New Orleans, USA
    • Rich culture, music, and culinary delights.
  4 Vietnam (Hanoi & Halong Bay)
    • Scenic landscapes and cooler temperatures.
  5 Amazon Rainforest (Brazil)
    • Vibrant wildlife and immersive nature experiences.

 If you have specific travel dates or any preferences regarding flights or accommodations, please
 let me know, and I can assist you further!

User:
Bye!
(gentenv) PS C:\Users\Guru\OneDrive\Desktop\Gentopia-Mason\GentPool> what is os
what : The term 'what' is not recognized as the name of a cmdlet, function, script file, or operable program. Check the spelling of the name, or if a path was included, verify that the path is correct and
try again.
At line:1 char:1
+ what is os
+ ~~~~
    + CategoryInfo          : ObjectNotFound: (what:String) [], CommandNotFoundException
    + FullyQualifiedErrorId : CommandNotFoundException

(gentenv) PS C:\Users\Guru\OneDrive\Desktop\Gentopia-Mason\GentPool> $env:OS
Windows_NT
(gentenv) PS C:\Users\Guru\OneDrive\Desktop\Gentopia-Mason\GentPool>
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